

TEAM XX



All Ports Tours

Cruise Line

Internal Security Assessment Report

All Ports Tours Cruise Line

11/2/2025

Version 7.23

Confidential

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2. Statement of Confidentiality and Liability

This assessment was conducted in accordance with the terms and conditions established by All Ports Tours Cruise Line, with all activities performed strictly within the authorized scope and procedures defined in the engagement overview. All findings and recommendations contained in this report are based on evaluations performed during the specified testing period.

Due to the limited timeframe and defined scope of the engagement, the findings presented herein should not be considered an exhaustive identification of all potential vulnerabilities or risks within All Ports Tours Cruise Line's network or systems. This report is provided solely for the purposes outlined in the engagement agreement and does not constitute legal, compliance, or regulatory advice.

Disclaimer:

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3. Contact Information

Client Information

Client	All Ports Tours
Primary Contact	Bailey Finn
Approvers	PERSON 1 PERSON 2

Note that all personnel indicated in the *Approvers* section are authorized to modify the scope and terms of engagement at their discretion.

SOCAL-XX Contact Information

Name	Team-XX Security Consulting
Address	
Phone	
Email	

Version Information

Version	Date	Description
1.0	11/01/2025	Initial Penetration Test for All Port Tours

Executive Summary

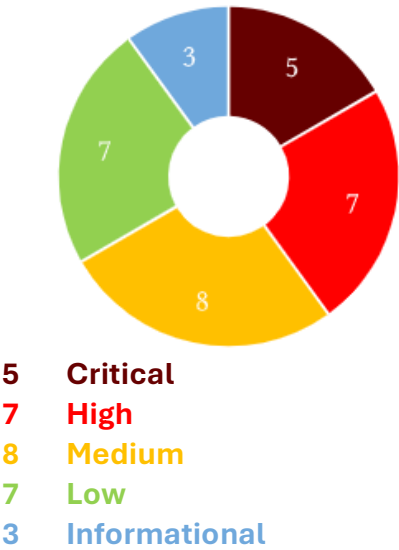
Throughout November 1st, **SOCAL-XX** conducted a security assessment on All Ports Tours Cruise Line’s (APTCL) network **SPECIFY MULTIPLE NETWORKS HERE IF NECESSARY**. The assessment aimed to evaluate APTCL’s adherence to the best security practices, compliance with common standards, as well as the overall safety and well-being of any individuals relying on the analyzed systems.

This assessment identified **NUMBER** vulnerabilities deemed as critical, with a total of **NUMBER** findings of all types identified in the scope of the APTCL network. APTCL faces a **critical** risk of compromise, and immediate attention to the network is necessary.

To enhance overall security measures, **SOCAL-XX** has created a comprehensive and holistic security strategy to address and resolve these issues as described in **Section X, Strategic Plan**. In general, **SOCAL-XX** recommends that APTCL follow industry best practices, such as keeping services and systems up-to-date with regular patches and updates, implementing strong encryption and authentication, and strengthening access controls.

Additionally, it is advisable for APTCL to implement employee training programs, focusing on password reuse and complexity, alongside regularly scheduled sessions addressing the awareness and prevention of social engineering attacks. Implementing proactive steps such as these will contribute to the strength of the overall security posture of All Ports Tours Cruise Line.

Findings



Timeline

Assessment began
2025-date-here

Assessment concluded
2025-date-here

Report submitted
2025-date-here

Scope

IP addresses, cloud stuff, phishing here

Compliance Review

During Team-XX's engagement and assessment of All Port Tours Cruise Line's system, there were several identified discrepancies from established Compliance Standards, including PCI DSS and BOAT COMPLIANCE LOOK UP LATER. Any deviation from these compliance standards can result in compromise of APTCL's cyber systems and result in severe impact in customer data confidentiality and integrity, as well as operational stability and availability during standard operations. If not immediately remedied, the potential consequences of these vulnerabilities include security intrusions, sensitive customer data and system data leakage, service disruptions and unavailability, leading to a loss in customer and stakeholder confidence in the APTCL's brand and serious legal repercussions and fines. Given the severity of the vulnerability findings, immediate attention and corrective action are strongly recommended to avoid these regulatory issues. Detailed insights into findings, recommended actions for remediation, and important insights are provided in the accompanying sections and Appendix (ALEX).

Boat Compliance for later

Payment Card Industry Data Industry Standard (PCI DSS)

PCI DSS is an information security standard used for the handling of credit card information from major card brands which controls requirements for control of cardholder data to reduce the likelihood of personal identifiable information of customers being exposed and combat credit card fraud. PCI DSS applies to any business entity that stores, processes, or transmits cardholder data. During the penetration testing engagement with the APTCL's system, Team-XX discovered a total of BIG NUMBER violations, that fail to comply with PCI DSS standards. These violations, if left unaddressed, can not only lead to the exposure of customer credit card information, but will ultimately lead to significant fines from the Payment Card Industry Security Standards Council. The exact cost of these fines varies depending on the severity, duration, and the volume of transactions within APTCL's non-compliant systems. It is critical for APTCL to address these security issues promptly to avoid the exposure of sensitive customer data and avoid severe legal penalties.

Breakdown of PCI DSS Violations

The following pie chart summarizes the violations by category to help All Port Tours Cruise Line prioritize mitigations.

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Engagement Overview

TEAM NAME was contracted by COMPANY to perform a security assessment on the security posture of Service. This engagement was conducted with the purpose of extensively testing the in-scope networks from the perspective of a threat actor attempting to gain access to unauthorized resources.

The phases of the engagement are as outlined:

1. External Enumeration

The engagement began with external reconnaissance and enumeration. Industry standard tools and techniques were used to identify open ports, running services, and potential attack vectors in the environment.

2. Initial Foothold

Using information gathered during initial enumeration, we identified and exploited vulnerabilities that allowed us to gain an initial foothold within the target environment, often as a low-level user.

3. Privilege Escalation

Local privilege escalation techniques were employed to elevate our access from low-level accounts to ones with administrative or root privileges. This step allowed for broader control over affected machines and increased access to internal resources.

4. Internal Enumeration and Pivoting

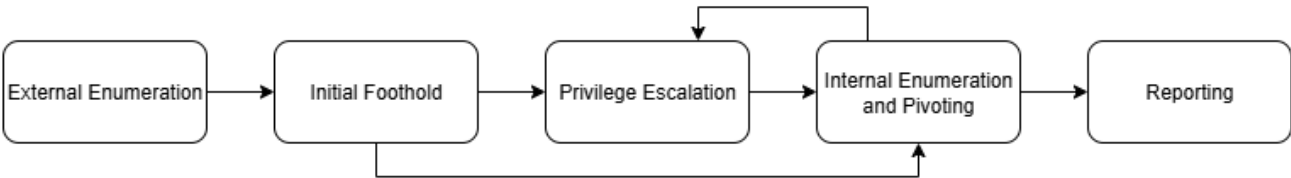
Following successful privilege escalation or initial footholds on high level users, we performed thorough internal network enumeration to map internal infrastructure and identify additional targets. Using our elevated access, we performed lateral movement across the network to compromise additional systems and expand our foothold. Techniques included BLANK.

5. Reporting

Upon completion of penetration testing, all findings were thoroughly documented and analyzed. This report presents a comprehensive overview of identified vulnerabilities. Each vulnerability finding includes a detailed technical descriptions, evidence of exploitation, risk ratings based on CVSS 4.0 scoring, and remediation

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recommendations. The report is structured to provide both executive-level summaries for strategic decision-making and technical details for security teams to implement effective countermeasures.



Scope of Evaluation:

Name	IP Range (CIDR) / Cloud Environment
XXXX Network	
XXXX Network	
XXX Network	

Provided Credentials:

User	Information

Engagement Timeline:

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Engagement Start:

Engagement End:

Report Delivered:

Attack Narrative

Day 1

Details

Day 2

Details

Network Topology

TEAM NAME conducted host discovery and enumeration across the defined in-scope subnets using industry-standard tools. Based on these findings, we constructed a network topology diagram to illustrate the structure of the **COMPANY** network.

Insert image of topology

Add pentesting methodology here

Strategic Remediation Plan

Throughout the assessment, Team-XX recognized multiple security controls implemented by APTCL. Team-XX recommends APTCL to continue upholding these controls to maintain its security posture. There are, however, key areas of improvement where additional security controls should be implemented. Refer to X.3 for a summary of this plan

X.1 Key Security Strengths

A. XXX

Details

B. XXX

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Details

C. XXX

Details

X.2 Key Improvement Areas

A. XXX

Details

B. XXX

Details

C. XXX

Details

X.3 Key Improvement Areas

Details

Diagram of Plan Here

Technical Risk Assessment Metrics

To capture the principal characteristics of a vulnerability and produce a numerical score reflecting its severity, **TEAM-XX** used the [Common Vulnerability Scoring System V4.0](#) (CVSS 4.0). CVSS scores are commonly used to provide a point of comparison between vulnerabilities and to help prioritize their remediation. Scores are calculated based on a formula released by the Forum of Incident Response and Security Teams (FIRST) that factors in a variety of metrics approximating ease of exploit, scope and impact. Scores range from 0 to 10, with 10 being the most severe.

[Common Vulnerability Scoring System V4.0](#) (CVSS 4.0)

Severity	Score	Explanation
Critical	9.0-10.0	Vulnerabilities in this category should be monitored closely by management and addressed as soon as possible.
High	7.0-8.9	Vulnerabilities in this category can have a significant impact on confidentiality, availability or integrity.
Medium	4.0-6.9	The exploit of vulnerabilities in this category might require additional data or vectors from the attacker to be successful.
Low	0.1-3.9	While the vulnerability is not severe, it poses a potential security risk and should be addressed.
Informational	0.0	Vulnerability should be remediated to meet industry best practices.

Team-XX has also implemented a methodology for assessing the likelihood of vulnerability exploitation. The measurement is categorized into three levels: High, Medium and Low. These levels, shown below in **table X**, provide a nuanced understanding of the probability a vulnerability is exploited, focusing primarily on ease of exploitation and existing security features.

Likelihood	Explanation
High	Exploitation is likely, the vulnerability is easily discoverable and does not require significant prior knowledge or other information to exploit.
Medium	Exploitation is expected to happen at times, however the exploit may require prior knowledge to leverage or is not easily discoverable.
Low	Exploitation is improbable due to the complexity of the exploit, the prior knowledge required of the attacker, and the existing security features.

Team-XXs comprehensive approach to risk assessment, combining both likelihood and severity as outlined below in Table X, provides a holistic view of cybersecurity threats. This integrated approach to vulnerability assessment aids in strategic resource allocation and improves communication with stakeholders. By prioritizing vulnerabilities based on both impact and likelihood, organizations can efficiently allocate resources to address the most significant threats.

Risk Matrix

Risk Matrix		Impact (CVSS Score)				
		Informational	Low	Medium	High	Critical
Likelihood	Low	Informational	Low	Low	Medium	High
	Medium	Informational	Low	Medium	High	Critical
	High	Informational	Medium	Medium	High	Critical

Technical Findings

Findings Count					
Severity Rating	Critical	High	Medium	Low	Informational
Vulnerabilities	#	#	#	#	#

Findings	Severity Level	Affected Scope
C-01:	Critical	
H-01:	High	
M-01:	Medium	
L-01:	Low	
I-01:	Informational	

Critical:

Matrix Calculation		CVSS Score	Risk
Impact			Critical
Likelihood			
CVSSv3.1 Attack Vector			
MITRE ATT&CK			

Affected Scope	
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High:

Matrix Calculation		CVSS Score	Risk
Impact			High
Likelihood			
CVSSv3.1 Attack Vector			
MITRE ATT&CK			
Affected Scope			

Medium:

Matrix Calculation		CVSS Score	Risk
Impact			Medium
Likelihood			
CVSSv3.1 Attack Vector			
MITRE ATT&CK			
Affected Scope			

Low:

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Matrix Calculation		CVSS Score	Risk
Impact			Low
Likelihood			
CVSSv3.1 Attack Vector			
MITRE ATT&CK			
Affected Scope			

Informational:

Matrix Calculation		CVSS Score	Risk
Impact			Informational
Likelihood			
CVSSv3.1 Attack Vector			
MITRE ATT&CK			
Affected Scope			

Findings Description:**Description****Business Impact****Confidential**

Steps to Reproduce

Remediation Strategies

References

Appendix

Appendix A – Tools Used

During the engagement, the following accounts' credentials were discovered, and the accounts were subsequently compromised. The credentials for these accounts should be immediately changed to prevent further unauthorized access to internal systems and accounts.

Username	Type/Privilege	Method of Compromise
Admin	Administrator	FTP Leakage
AndyG	User	Brute-forceable password

Appendix B – Tools Used

Tool	Description
Metasploit	A modular exploitation and post-exploitation framework used to run common exploits against remote systems.
Impacket	A collection of Python classes and scripts that implement low-level network protocols (SMB, MS-RPC, Kerberos, etc.) used for exploiting Windows networking services.
Nmap	A fast, flexible network scanner for discovering hosts, open ports, and basic service/version information; frequently used for initial reconnaissance and service enumeration.
Burp Suite Community Edition	Local Web Proxy for inspecting and manipulating HTTP(S) requests. Allows for intercepting, manipulating, and crafting malicious requests to web applications.

Netexec	A lightweight remote command execution utility used to run commands on a remote host over a simple TCP connection; useful for quick interactive access in lab or controlled engagement environments. Compatible for enumeration and exploitation of SMB, LDAP, WINRM, and other Windows protocols.
Gobuster	A directory, DNS and virtual host brute-forcing tool written in Go, used to discover hidden files, directories, and subdomains on web servers by wordlist enumeration.
Enum4linux-ng	A tool for enumerating information from Windows (shares, users, groups, policies) via SMB/RPC protocol interactions; useful for gathering host and domain details during network reconnaissance.
FFUF	A fast web fuzzer for discovering hidden resources, parameters, and inputs by substituting words from wordlists into URL paths and request components.
Hashcat	A high-performance password cracking tool that performs dictionary, rule-based, and brute-force attacks against password hashes.
Hydra	A parallelized network login cracker supporting many protocols (SSH, RDP, HTTP, SMB, etc.) used for rapid credential guessing/testing during credential-validation phases.
John	A versatile password cracking tool that supports many hash formats and cracking modes; often used for offline password hash cracking.
Kerbrute	A fast Kerberos username enumeration and password spraying tool for Active Directory environments; typically used to identify valid accounts and test weak Kerberos credentials.

Wfuzz	A flexible web application fuzzer for brute-forcing GET/POST parameters, testing hidden resources, and probing for injection points using customizable payloads.
GoPhish	An open-source phishing campaign framework for creating, sending, and tracking phishing emails and landing pages in red-team or security awareness exercises.
Maltego	A graphical link-analysis tool for OSINT that maps relationships between people, domains, infrastructure, and other entities to visualize attack surfaces and gather external intelligence.
Aquatone	A reconnaissance tool that takes screenshots of web hosts across a domain (usually from subdomain enumeration) to quickly identify interesting web assets and potential targets.
SQLMap	An automated tool for detecting and exploiting SQL injection vulnerabilities and for performing database takeover and data extraction in a controlled testing context.
PEASS-NG	A set of local privilege escalation enumeration scripts (Windows/Linux) that automate discovery of common misconfigurations and vectors to help identify potential privilege escalation paths.
BloodHound	A graph-based Active Directory analysis tool that models permissions and relationships to find attack paths (e.g., privilege escalation and lateral movement) in AD environments.
Ligolo-ng	A tunneling tool that creates TCP/UDP tunnels over an encrypted channel, often used to proxy traffic through a compromised host for pivoting in segmented networks.

Certipy	A tool for auditing and abusing Active Directory certificate services (AD CS); helps identify certificate-based attack vectors and perform certificate-related reconnaissance.
Mimikatz	A post-exploitation tool for extracting plaintext credentials, Kerberos tickets, and credential artifacts from Windows memory and secure stores; commonly used to demonstrate credential exposure risks.
Rubeus	A C# toolset for interacting with Kerberos on Windows (requesting/renewing tickets, abusing service tickets) used in Kerberos-centric assessments and AD attacks.
BloodyAD	A collection/automation of offensive AD techniques (enumeration, abuse, exploitation) that wraps multiple capabilities useful for Active Directory assessments.
SWAKS	A versatile command-line SMTP testing tool for crafting and sending customized email messages, useful for testing mail server behavior and phishing infrastructure.
LaZagne	A credentials recovery tool that searches for a local machine for stored credentials from many applications (browsers, mail clients, databases) to assess persistence of secrets on endpoints.
Evil-WinRM	A PowerShell Remoting (WinRM) client designed for red teamers that provides an interactive shell and file transfer for Windows hosts where WinRM connectivity is available.
LDAPSearch	A command-line LDAP query utility for retrieving directory objects, attributes, and schema; used to enumerate users, groups, service accounts, and metadata during Active Directory reconnaissance.
CyberChef	A browser-based “recipe” tool for decoding, encoding, transforming, and parsing data (hashes, tokens, logs);

	used for rapid on-the-fly analysis and format conversion during triage and exploitation workflows.
Responder	A network responder/poisoning framework for LLMNR/NetBIOS/mDNS that captures authentication attempts (NTLM/Net-NTLMv2) and service requests.

Appendix C - Methodology

TODO

- Pentesting standards
 - o http://www.pentest-standard.org/index.php/Main_Page
- OSINT
- Phishing
 - o <https://www.nist.gov/publications/nist-phish-scale-user-guide>

OWASP Top 10

OWASP Top 10 is a standard web application security framework that is a representation of the most critical and commonly seen security risks pertaining to web applications. This table highlights the critical vulnerabilities present within the web applications in APT's infrastructure.

Vulnerability	Test Results
Broken Access Control	FAIL
Cryptographic Failure	FAIL
Injection Insecure Design	FAIL FAIL
Security Misconfiguration	FAIL
Vulnerable and Outdated Component	PASS
Identification and Authentication Failures	PASS

Software and Data Integrity Failures	PASS
Security Logging and Monitoring Failures	PASS
Server-Side Request Forgery	PASS

Appendix D – CVSS 4.0

Team-XX utilized the Common Vulnerability Scoring System (CVSS) during the engagement to classify the severity level of each discovered vulnerability. CVSS is a common, industry standard scale that calculates the severity of a vulnerability based on important factors including attack complexity, scope, permissions required, subsequent system access, and impact on confidentiality, integrity, and accessibility. CVSS 4.0, implemented in 2023, was utilized as it is the most modern implementation of the scoring system. All findings were reported on a rating from (0.0-10.0) with the relevant CVSS scoring calculations linked for ease of use.

Appendix E – MITRE ATT&CK Framework

The MITRE ATT&CK Framework is a global knowledge base of real-world adversary tactics and techniques. The ATT&CK knowledge base is used for a foundation of specific threat modeling, to allow defensive security teams to better understand and stop advanced threats and cyber-attacks. MITRE provides specific remediations for all tactics and techniques to inform defensive security operations teams and the best ways to defend against adversary threats. Relevant MITRE techniques and remediations were provided for all findings to better inform ATP on how each finding could be exploited and remediated.

Appendix F – Artifacts

During the course of the engagement, **Team-XX** modified parts of APT's network to aid in vulnerability discovery. All artifacts were removed before the end of the engagement to ensure the network attack surface was not further increased through the processes performed in the engagement.